

Computability & Logic – Exam June 2022

ID

Name

You have 3 hours to finish this exam. **Note:** this is an open-book exam.

You may use books, lecture slides and other notes on paper.

The use of electronic devices is *not* permitted (dispensations may apply).

Each of the following 30 questions has 4 possible answers of which exactly one is correct. For each question, you may select one or more answers by checking the appropriate boxes. Your test is graded as follows:

- If you select only the correct answer, you receive 2 points.
- If you select 2 answers, one of which is correct, you receive 1 point.
- If you select 3 answers, one of which is correct, you receive 0.41 points.
- If you select no answer or all answers, you receive 0 points.
- If you select only one answer, and it is wrong, you receive -0.67 points.
- If you select 2 answers, that are both wrong, you receive -1 point.
- If you select 3 answers, that are all wrong, you receive -1.25 points.

Note that perfect answers yield a score of 60 and random guessing may give you a negative score. For the logical basis of this score system, see Frandsen and Schwartzbach, *A singular choice for multiple choice*.

Success!

Example

Regular Languages

Question 1

Consider the regular expression $E := (a^*b + b^*c)^*$. Which one of the following words is *not* a member of the language $\mathcal{L}(E)$?

- a ☐ $abac$
- b ☐ abc
- c ☐ $abbc$
- d ☐ $cbab$

Question 2

Let $\mathcal{L}_1$, $\mathcal{L}_2$ and $\mathcal{L}_3$ be languages over the same alphabet Σ .

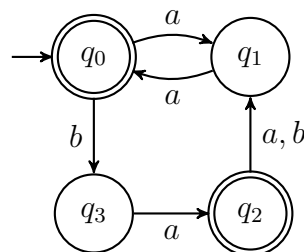
Assume that $\mathcal{L}_1 \subseteq \mathcal{L}_2 \subseteq \mathcal{L}_3$. Assume that $\mathcal{L}_1$ and $\mathcal{L}_3$ are regular.

Which of the following statements does *not* follow from the assumptions above?

- a ☐ $\mathcal{L}_2$ is regular.
- b ☐ If $\mathcal{L}_3$ is finite then $\mathcal{L}_2$ is regular.
- c ☐ If $\mathcal{L}_2$ is regular then $\mathcal{L}_2' \cap \mathcal{L}_3$ is regular.
- d ☐ $\mathcal{L}_1 \cap \mathcal{L}_2$ is regular.

Question 3

Consider the following *incomplete* DFA M :



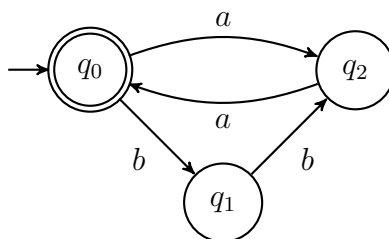
We compute the (unique) minimal *complete* DFA M' , such that $\mathcal{L}(M) = \mathcal{L}(M')$.

What is the number of states of M' ?

- a ☐ 3 states
- b ☐ 2 states
- c ☐ 4 states
- d ☐ 5 states

Question 4

Consider the following incomplete DFA M :

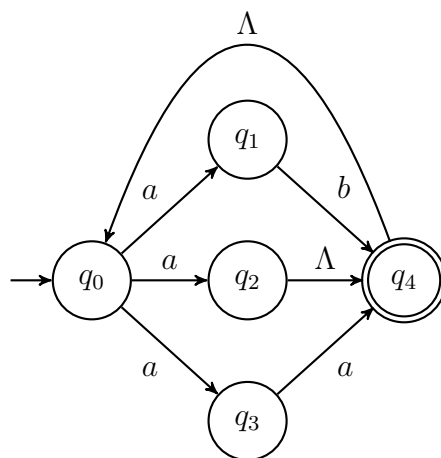


Which of the regular expressions below does *not* describe the language $\mathcal{L}(M)$?

- a ☐ $(aa)^* + (bba)^*$
- b ☐ $(aa + bba)^*$
- c ☐ $((a + bb)a)^*$
- d ☐ $(aa + bba + aabba)^*$

Question 5

Consider the following NFA M :



We compute the minimal complete DFA M' , such that $\mathcal{L}(M) = \mathcal{L}(M')$.

What is the number of (accepting) states in M' ?

- a ☐ M' has 4 states, of which 2 states are accepting.
- b ☐ M' has 4 states, of which 1 state is accepting.
- c ☐ M' has 3 states, of which 1 state is accepting.
- d ☐ M' has 3 states, of which 2 states are accepting.

Question 6

We define the following three languages over the alphabet $\Sigma = \{a, b\}$.

Recall that $\#_a(w)$ is defined as the number of a 's that occur in word w .

- $\mathcal{L}_1 = \{w \in \Sigma^* \mid \#_a(w) \text{ is a multiple of } 5\}$
- $\mathcal{L}_2 = \{w \in \Sigma^* \mid \#_a(w) \text{ is a square number}\}$
- $\mathcal{L}_3 = \{w \in \Sigma^* \mid \#_a(w) = \#_b(w)\}$

Which of these languages is *regular*?

- a ☐ Only $\mathcal{L}_1$ is regular
 b ☐ None of these languages is regular
 c ☐ $\mathcal{L}_1$ and $\mathcal{L}_2$ are regular, but $\mathcal{L}_3$ is not regular
 d ☐ $\mathcal{L}_1$ and $\mathcal{L}_3$ are regular, but $\mathcal{L}_2$ is not regular
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Example**Context-Free Languages**

Question 7

Consider the following Context-Free Grammar:

$$G = \begin{cases} S & \rightarrow AB \\ A & \rightarrow aA \mid B \\ B & \rightarrow bB \mid b \end{cases}$$

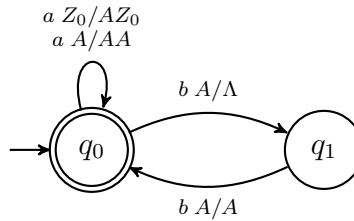
G happens to be an ambiguous grammar.

Which of the following words has more than one derivation tree in G ?

- a ☐ The word $aaabbb$
 b ☐ The word $aaabb$
 c ☐ The word $aabb$
 d ☐ The word ab

Question 8

Consider the following PDA (Push-Down Automaton).



Which of the following words is *not* accepted by final state in this PDA?

- a ☐ The word *abb*
- b ☐ The word *a*
- c ☐ The word *aabb*
- d ☐ The word *aaabbbb*

Question 9

Consider the following grammar in Chomsky Normal Form.

$$G = \begin{cases} S & \rightarrow AB \\ A & \rightarrow AB \mid a \\ B & \rightarrow BX \mid b \\ X & \rightarrow b \end{cases}$$

We apply the CYK-parsing algorithm to the input word *abbb*. The matrix is already partially filled.

1	{A}	..	..	??
2	—	{X, B}	..	..
3	—	—	{X, B}	..
4	—	—	—	{X, B}
	1	2	3	4

Which set should be filled in at the top-right position?

- a ☐ The set $\{A, S\}$
- b ☐ The set $\{S\}$
- c ☐ The set $\{B, S\}$
- d ☐ The set $\{A, B, S\}$

Question 10

All of the following languages are Context-Free, except one.

Which language is *not* context-free?

- a ☐ $\{a^m b^n c^p \mid m = n = p\}$
- b ☐ $\{a^m b^n c^p \mid m = n\}$
- c ☐ $\{a^m b^n c^p \mid m = p\}$
- d ☐ $\{a^m b^n c^p \mid m + n = p\}$

Question 11

Which of the following statements is true?

- ☐ a Every Regular Language is Deterministic Context-Free.
- ☐ b Every Deterministic Context-Free Language is Regular.
- ☐ c Every Context-Free Language is Deterministic Context-Free).
- ☐ d The intersection of a Regular Language and a Context-Free Language is always Regular.

Question 12

Assume that L_1 is a Regular Language and L_2 and L_3 are Context-Free Languages.
(recall that L' denotes the complement of L)

Which of the following statements follows with certainty?

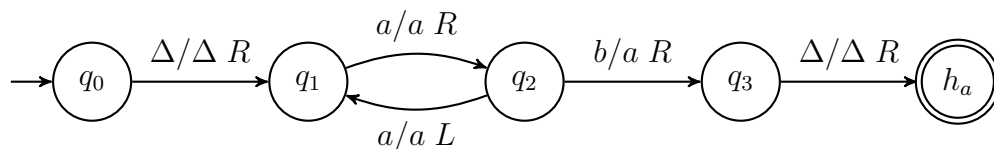
- ☐ a $L'_1 \cup L_2$ and $L_1 \cap L_2$ are both Context-Free
- ☐ b $L_1 \cup L'_2$ and $L_1 \cap L_2$ are both Context-Free
- ☐ c $L'_1 \cup L_2$ and $L_2 \cap L_3$ are both Context-Free
- ☐ d $L_1 \cup L'_2$ and $L_2 \cup L_3$ are both Context-Free

Example

Turing Machines and Decidability

Question 13

Consider the following 1-tape Turing Machine on input alphabet $\Sigma = \{a, b\}$:



Which statement on this TM is true?

- ☐ a This TM accepts the language $\{ab\}$ but does not decide it.
- ☐ b This TM decides the language $\{ab\}$.
- ☐ c This TM accepts the language $\{aa\}$ but does not decide it.
- ☐ d This TM decides the language $\{aa\}$.

Question 14

We define the following sets:

- $A = \{1, \dots, 100\}$
- $B = A^2 = A \times A$
- $C = \mathbb{N}$
- $D = \mathbb{N}^2 = \mathbb{N} \times \mathbb{N}$
- $E = 2^{\mathbb{N}} = \mathbb{N} \rightarrow \{0, 1\}$

We order these sets according to their cardinality. What is the proper order?
(we write $=$ for same cardinality and $<$ for strictly smaller cardinality)

- a ☐ $A < B < C = D < E$
- b ☐ $A < B < C < D = E$
- c ☐ $A = B < C = D < E$
- d ☐ $A = B < C = D = E$

Question 15

If L is a *recursive* language, then:

- a ☐ There exists a deterministic TM that accepts L and terminates on all input.
- b ☐ All Turing Machines that accept L terminate on all input.
- c ☐ L is recognized by a (possibly non-deterministic) Push Down Automaton.
- d ☐ L' (=complement) cannot be a recursive language.

Question 16

If L is a *recursively enumerable* language, then:

- a ☐ There exists a deterministic TM that accepts L and terminates on all $w \in L$.
- b ☐ There exists a deterministic TM that accepts L and terminates on all input.
- c ☐ L' (=complement) must also be a recursively enumerable language.
- d ☐ L' (=complement) cannot be a recursively enumerable language.

Question 17

Which statement is wrong?

- a ☐ If L_1 and L_2 are context-free, then so are $L_1 \cup L_2$ and $L_1 \cap L_2$.
- b ☐ If L_1 and L_2 are recursive, then so are $L_1 \cup L_2$ and $L_1 \cap L_2$.
- c ☐ If L_1 and L_2 are recursively enumerable, then so are $L_1 \cup L_2$ and $L_1 \cap L_2$.
- d ☐ If L_1 and L_2 are regular, then so are $L_1 \cup L_2$ and $L_1 \cap L_2$.

Question 18

Consider the following decision problems on Turing Machines:

- *Accept- Λ* : Input TM M_1 . Question: $\Lambda \in \mathcal{L}(M_1)$?
- *Equivalent*: Input TM M_2, M_3 . Question: $\mathcal{L}(M_2) = \mathcal{L}(M_3)$?

We may use the following simple Turing Machines as auxiliary components:

- M_Σ : a TM that accepts all words in Σ^* .
- *Write*(w): a TM that just writes word w on tape and rewinds.
- *Erase*: a TM that erases all input from the tape.

Which of the following problem reductions can be used to show that *Accept- Λ* $\leq$ *Equivalent*? (recall that $\rightarrow$ denotes composition of TMs)

- ☐ a $F(M_1) := (\text{Erase} \rightarrow M_1, M_\Sigma)$
- ☐ b $F(M_1) := (\text{Write}(\Lambda) \rightarrow M_1, M_\Sigma)$
- ☐ c $F(M_2, M_3) := \text{Write}(\Lambda) \rightarrow M_2 \rightarrow M_3$
- ☐ d $F(M_2, M_3) := \text{Erase} \rightarrow M_2 \rightarrow M_3$

Question 19

Which of the following problems satisfies the two conditions of Rice's theorem?

- ☐ a Given a TM M , is the word Λ accepted by M ?
- ☐ b Given a TM M , does M have more than 100 states?
- ☐ c Given a TM M , can M perform more than 100 steps on input Λ ?
- ☐ d Given a TM M , is the language accepted by M countable?

Question 20

Consider the following instance of Post's Correspondence Problem (PCP) with the three pieces $\{(123, 12), (2, 1312), (131, 31)\}$. The question is if this is a yes-instance (this instance can be solved).

123	2	131
12	1312	31

- ☐ a This instance is solvable, but at least one piece must be reused
- ☐ b This instance is solvable, even without reusing any pieces
- ☐ c This instance is not solvable at all, but other PCP instances can be solved
- ☐ d This instance is not solvable, and no PCP instances can be solved

Question 21

Which of the following problems on Context-Free Grammars is *undecidable*?

(we write L' for the complement of L)

- ☐ a Given input CFG G_1 and G_2 and word w . Question: $w \in \mathcal{L}(G_1)' \cup \mathcal{L}(G_2)'$?
 - ☐ b Given input CFG G and word w . Question: $w \in \mathcal{L}(G)$?
 - ☐ c Given input CFG G_1 and G_2 and word w . Question: $w \in \mathcal{L}(G_1) \cup \mathcal{L}(G_2)$?
 - ☐ d Given input CFG G and Regular Expression E . Question: $\mathcal{L}(G) \subseteq \mathcal{L}(E)$?
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Question 22

Which of the following statements is wrong?

- ☐ a If some problem in NP can be solved in polynomial time, then SAT is in P.
 - ☐ b If SAT can be solved in polynomial time, then every NP-complete problem is in P.
 - ☐ c If SAT can be solved in polynomial time, then every problem in NP is in P.
 - ☐ d If some NP-complete problem can be solved in polynomial time, then SAT is in P.
-

Example**Logic**

Question 23

What is the truth table of the propositional logic formula $(p \rightarrow q) \rightarrow (p \rightarrow \neg q)$?

p	q	$(p \rightarrow q) \rightarrow (p \rightarrow \neg q)$
T	T	F
T	F	T
F	T	T
F	F	T

p	q	$(p \rightarrow q) \rightarrow (p \rightarrow \neg q)$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$(p \rightarrow q) \rightarrow (p \rightarrow \neg q)$
T	T	F
T	F	F
F	T	T
F	F	T

p	q	$(p \rightarrow q) \rightarrow (p \rightarrow \neg q)$
T	T	F
T	F	F
F	T	F
F	F	F

Question 24

Consider the following formula in propositional logic: $(p \rightarrow \neg p) \rightarrow \neg p$.

This formula is:

- a ☐ A tautology.
 - b ☐ Satisfiable, but not valid.
 - c ☐ Unsatisfiable.
 - d ☐ None of the above.
-

Question 25

Consider the list of proof rules of natural deduction for proposition logic:

$(\text{premise/assumption}), \wedge i, \wedge e_1, \wedge e_2, \vee e, \vee i_1, \vee i_2, \rightarrow i, \rightarrow e, \neg i, \neg e, \neg\neg e, \perp e$.

The sequent " $(\neg p \rightarrow q) \vdash (\neg q \rightarrow p)$ " can be proved by the following box-style proof, where we replaced the names of the applied rules by question marks:

1.	$\neg p \rightarrow q$	(premise)
2.	$\neg q$	(assumption)
3.	$\neg p$	(assumption)
4.	q	???
5.	$\perp$	???
6.	$\neg\neg p$	???
7.	p	???
8.	$\neg q \rightarrow p$	???

Which rules fit on the five question marks in the proof? (in the correct order)

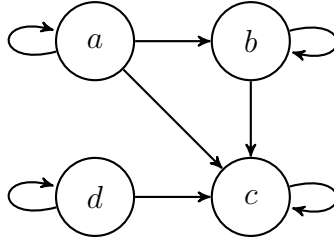
- a ☐ $\rightarrow e, \neg e, \neg i, \neg\neg e, \rightarrow i$
- b ☐ $\rightarrow i, \neg e, \neg i, \neg\neg e, \rightarrow e$
- c ☐ $\rightarrow e, \neg i, \perp e, \neg\neg e, \rightarrow i$
- d ☐ $\neg e, \perp i, \neg i, \neg\neg e, \rightarrow i$

Question 26

Consider the following model of predicate logic, where the vocabulary contains a binary relation R . The model consists of a universe $\{a, b, c, d\}$ and the relation

$$R = \{(a, a), (b, b), (c, c), (d, d), (a, b), (a, c), (b, c), (d, c)\}$$

The model can be visualized as follows:



Which one of the following formulas in predicate logic does *not* hold in this model?

- a ☐ $\forall x \forall y \exists z (R(z, x) \wedge R(z, y))$
- b ☐ $\forall x \forall y \exists z (R(x, z) \wedge R(y, z))$
- c ☐ $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$
- d ☐ $\exists z \forall x R(x, z)$

Question 27

Consider the following proof rules for Natural Deduction (without equality and negation):

$$(premise/assumption), \wedge i, \wedge e_1, \wedge e_2, \vee e, \vee i_1, \vee i_2, \rightarrow i, \rightarrow e, \forall i, \forall e, \exists i, \exists e.$$

Consider the following sequent in Predicate Logic (a and b are constants):

$$R(b) \rightarrow Q(b), \quad \forall x (P(x) \rightarrow (Q(a) \vee R(x))) \quad \vdash \quad P(b) \rightarrow \exists x Q(x)$$

In the statements below, we disregard applications of premise/assumption, and the mentioned rules can be applied any number of times in any order.

Which statement is true?

- a ☐ This sequent can be proved using rules $\rightarrow e, \rightarrow i, \vee e, \forall e, \exists i$ (in any order)
- b ☐ This sequent is not a theorem, so it cannot be proved in natural deduction.
- c ☐ This sequent can be proved using $\rightarrow e, \rightarrow i, \vee e, \exists e, \forall i$ (in any order)
- d ☐ This sequent can be proved using $\rightarrow e, \rightarrow i, \vee i_1, \vee i_2, \forall e, \exists e$ (in any order)

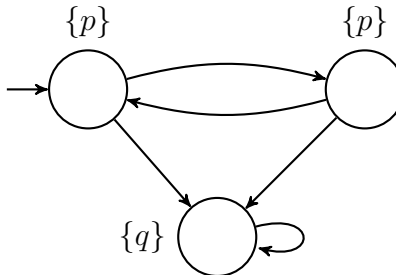
Question 28

Which one of the following statements is *wrong*?

- a ☐ If $\neg\phi$ is not satisfiable in Arithmetic, then ϕ is provable in Peano Arithmetic.
- b ☐ If $\neg\phi$ is not satisfiable in Predicate Logic, then ϕ is provable in Natural Deduction.
- c ☐ If $\neg\phi$ is satisfiable in Predicate Logic, then ϕ is not provable in Natural Deduction.
- d ☐ If $\neg\phi$ is satisfiable in Arithmetic, then ϕ is not provable in Peano Arithmetic.

Question 29

Consider the following Kripke structure:



Which one of the following LTL formulas holds for this Kripke structure?

- a ☐ $p \mathbf{W} q$
- b ☐ $p \mathbf{U} q$
- c ☐ $\mathbf{F} \mathbf{G} q$
- d ☐ $\mathbf{G} (q \rightarrow p)$

Question 30

Consider the following program P , on integer variables x, y, z :

$$\begin{aligned} z &:= x + 1; \\ x &:= y + 1; \\ y &:= z + 1; \end{aligned}$$

Only one of the following Hoare triples about P is *incorrect*. Which one?

- a ☐ $\{x > 10 \wedge y < 5\} P \{y > 10 \wedge x < 5\}$
- b ☐ $\{x > 8 \wedge y < 4\} P \{y > 10 \wedge x < 5\}$
- c ☐ $\{x > 10 \wedge y < 4\} P \{y > 10 \wedge x < 5\}$
- d ☐ $\{x > 8 \wedge y < 3\} P \{y > 10 \wedge x < 5\}$